UPDATED TECHNICAL CORRECTION PACKAGE

Lanmodulin Dy-selectivity workflow

Corrected figures, tables, reviewer-response map, and validated OpenMolcas prototype evidence

Manuscript: JMGM-D-26-01758

Scope: Technical corrections and prototype evidence; main manuscript intentionally unchanged

Repository: https://github.com/RahatulAshakin/lanm-dy-selectivity

Audit trail: original audit 549b22918e1edf409e0c75831cb9af7069468275; update prepared from main e425926d3b22de50beef9dda92a657767b52df66

Updated: 2026-08-16

Targeted result after correction

The technically supportable result is a model-prioritized Hans pocket candidate for additional quantum and experimental testing. The present repository does not establish thermodynamic binding free energy, Dy selectivity, zero-waste separation, or a validated quantum-state mechanism.

Area	Status	Submission use
Classical data correction	Ready	Corrected masks, funnel, lineage, model settings, OpenMM replica data, and dimensionless model score
Quantum prototype execution	Validated	Genuine OpenMolcas H2 reference and atomic Dy CAS(9,7)/SOC/SINGLE_ANISO runs completed; these are not LanM results
Manuscript text	Unchanged	Author name, title/abstract claims, citations, and prose edits are recorded but not applied
Repository remediation	Included	Updated report/status tables plus prototype inputs, logs, figures, tables, provenance, and selected source artifacts

Classical corrections trace to original audit commit 549b22918e1edf409e0c75831cb9af7069468275; this update was prepared from main commit e425926d3b22de50beef9dda92a657767b52df66.

1. Scope and correction decision

This package responds to the supplied reviewer comments through technical and reproducibility corrections only. It does not alter the manuscript. The package separates supported corrections from author-dependent work so that unsupported quantum values, mechanisms, or experimental claims are not introduced merely to satisfy a review point.

1.1 What can be corrected now

Correction	Concrete update
Metric terminology	Remove the energy-like transformation and report a dimensionless Composite Quality Index.
Design accounting	Reconcile 70 role memberships to 66 unique positions and separate positions from generated sequences.
Sampling transparency	Report exact MPNN temperatures, seeds, counts, and shortlist units.
Rosetta description	Replace structural-realism wording with one-structure score-only triage and explicit limitations.
OpenMM reporting	Expose all 45 replica/metal/candidate means, parameter values, seeds, duration, metric definition, and force-field caveats.
Structure scope	Distinguish deposited A-D chains from author/PISA A2 assemblies and clarify chain-A versus multichain computational contexts.

Full machine-readable tables are included in the tables/ directory.

1.2 What cannot be corrected without author data

The project repository still contains no project-specific Dy/Nd ORCA or OpenMolcas production inputs and outputs for the LanM clusters. This update adds genuine OpenMolcas reference and atomic-Dy prototype files, including natural occupations, SOC energies, orbital artifacts, and independent repeat evidence. Because the prototype is not a LanM cluster and includes no Nd production calculation, it does not regenerate or validate manuscript Figures 6-7.

2. Reviewer-response correction matrix

The matrix below remains the point-by-point technical response. Prototype evidence is marked separately from project-specific evidence; a prototype result never changes a project-data gap to complete.

Item	Reviewer issue	Technical correction	Status
R1.1 / R2-B	Undefined and thermodynamically invalid pocket delta-G proxy	Replace with dimensionless Composite Quality Index = 0.7 ligand confidence + 0.3 overall confidence; remove -RT ln transformation. Evidence: Figure TC5; model_quality_index.csv.	Corrected
R1.2	Natural occupations and active-space suitability	Add exact seven-orbital natural occupations for every root of the official atomic Dy CAS(9,7) prototype and preserve source logs. Project-specific Dy/Nd LanM occupations and active-space justification remain required.	Partially addressed: prototype only
R1.3	CASPT2 shift, intruder states, number of states	Add a validated H2 CASPT2 reference energy check. Project-specific Dy/Nd CASPT2 state counts, shifts, intruder diagnostics, energies, and convergence remain required.	Partially addressed: reference only
R1.4	Reduced-cluster size and environmental effect	Require atom/residue inventory, total charge/multiplicity, caps/constraints, coordinates, and sensitivity comparison. Evidence: quantum_evidence_gap_register.csv.	Blocked: source data absent

Item	Reviewer issue	Technical correction	Status
R1.5	Zero-waste claim lacks experimental evidence	Technical package states that no experimental separation or waste-balance evidence is present and narrows the result to a computational candidate-prioritization claim. Evidence: Technical report Sections 1 and 6.	Flagged; manuscript wording outside scope
R1.6	Only Al and Fe competitors	State that Al(III)/Fe(III) were the tested off-target subset. Do not generalize to Ca/Mg/Zn; recommend an expanded panel. Evidence: modeling_specification.csv.	Corrected limitation
R1.7	MPNN parameters and filtering unspecified	Expose exact temperatures, seeds, generated/unique/retained counts, and selection units. Evidence: Figure TC2; sequence_funnel.csv; modeling_specification.csv.	Corrected
R1.8 / R2-S4	Rosetta realism criteria/cutoffs unspecified	Correct description to score-only triage: score_jd2, nstruct=1, fixed seed, no relax, no acceptance cutoff, within-topology ranking only. Evidence: modeling_specification.csv.	Corrected and qualified
R1.9	Hans pocket versus interface unclear	Distinguish chain-A pocket model from chain-A-designed/B-D-fixed crystallographic context; attribute score difference to model confidence, not inferred structural causality. Evidence: Figure TC3; finalist_lineage.csv.	Corrected
R2-A1	Spin-orbit coupling not documented	Add genuine atomic-Dy RASSI/SOC and SINGLE_ANISO outputs with 66 states and 33 Kramers pairs. Project-specific LanM SOC treatment and state ordering remain required.	Partially addressed: prototype only
R2-A2	Relativistic Hamiltonian and basis sets absent	Record OpenMolcas v26.06, DKH/AMFI, ANO-RCC contraction, RICD/CDTH settings for the atomic-Dy prototype. Project-specific Hamiltonian/ECP and basis definitions remain required.	Partially addressed: prototype only
R2-A3	Active orbitals/occupations across roots not demonstrated	Add prototype occupation tables, RasOrb and Molden source artifacts. These demonstrate the extraction path but do not establish the LanM active space or replace Figure 7.	Partially addressed: prototype only
R2-C1	Metal force-field provenance and Al/Fe artifacts	Report actual generic OPC3-compatible 12-6-4 parameters; state tuned chelator values are absent; treat Al/Fe distances as model-dependent geometry, not affinity evidence. Evidence: metal_parameter_values.csv; Figure TC4.	Corrected with caution
R2-Minor 1	Christopher Stuetzle spelling	Record the requested spelling correction in the response matrix only; manuscript author line is intentionally unchanged in this package. Evidence: reviewer_response_matrix.csv.	Deferred: manuscript outside scope
R2-Minor 2	Arg100 not labeled	Identify chain-A Arg100 explicitly and show when it becomes Lys100 in the Hans pocket seed. Evidence: Figure TC3; finalist_lineage.csv.	Corrected
R2-Minor 3	References 30-36 irrelevant	Record as a manuscript bibliography action; no bibliography is modified in this technical-only package. Evidence: reviewer_response_matrix.csv.	Deferred: manuscript outside scope
R2-S4 Y-distance	Y distances converge near 2.19 Å	Report all nine replica values. Similarity is due to the common Y(III) model and near-identical local coordination; displayed equality was rounding, not duplicated data. Evidence: Figure TC4; openmm_replicates.csv.	Corrected
R2-Data	Deposit raw ORCA .in/.out	Deposit genuine OpenMolcas prototype inputs, successful console logs, extracted tables, plots, provenance, and selected orbital/anisotropy sources. Original project-specific ORCA/LanM files remain absent.	Partially addressed: project files required

Machine-readable version: tables/reviewer_response_matrix.csv.

3. Corrected figures

Figure TC1. Corrected design-mask accounting. Role categories overlap: 70 memberships correspond to 66 unique residue positions, with four one-count overlaps.

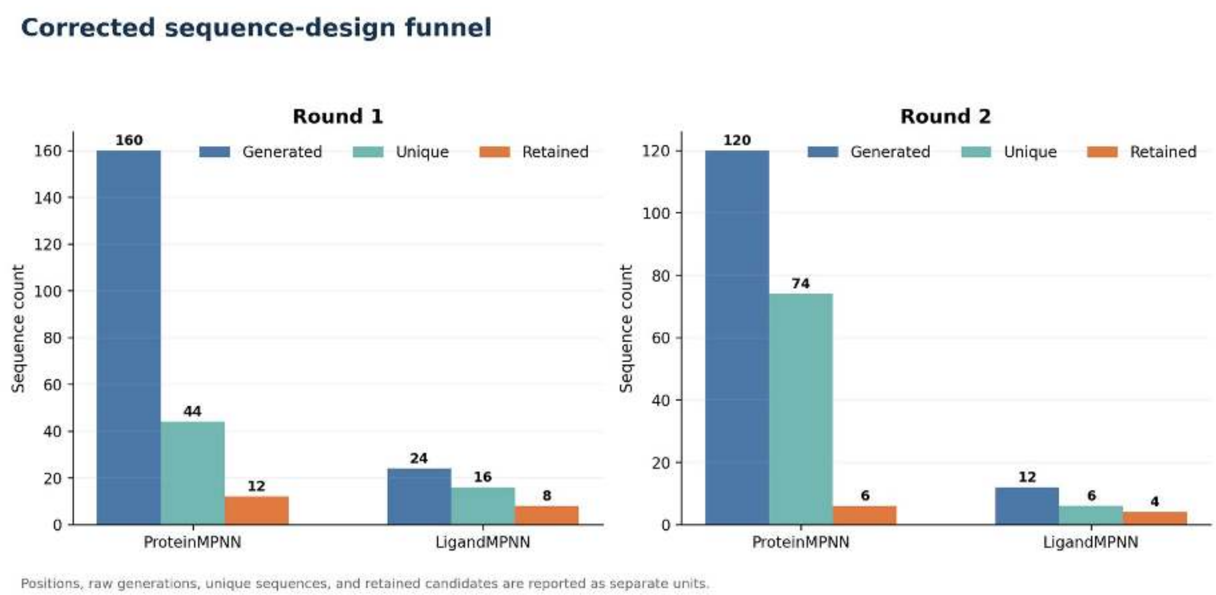

Figure TC2. Corrected design funnel. Raw generations, unique sequences, shortlisted sequences, scored candidates, and final OpenMM finalists are reported separately.

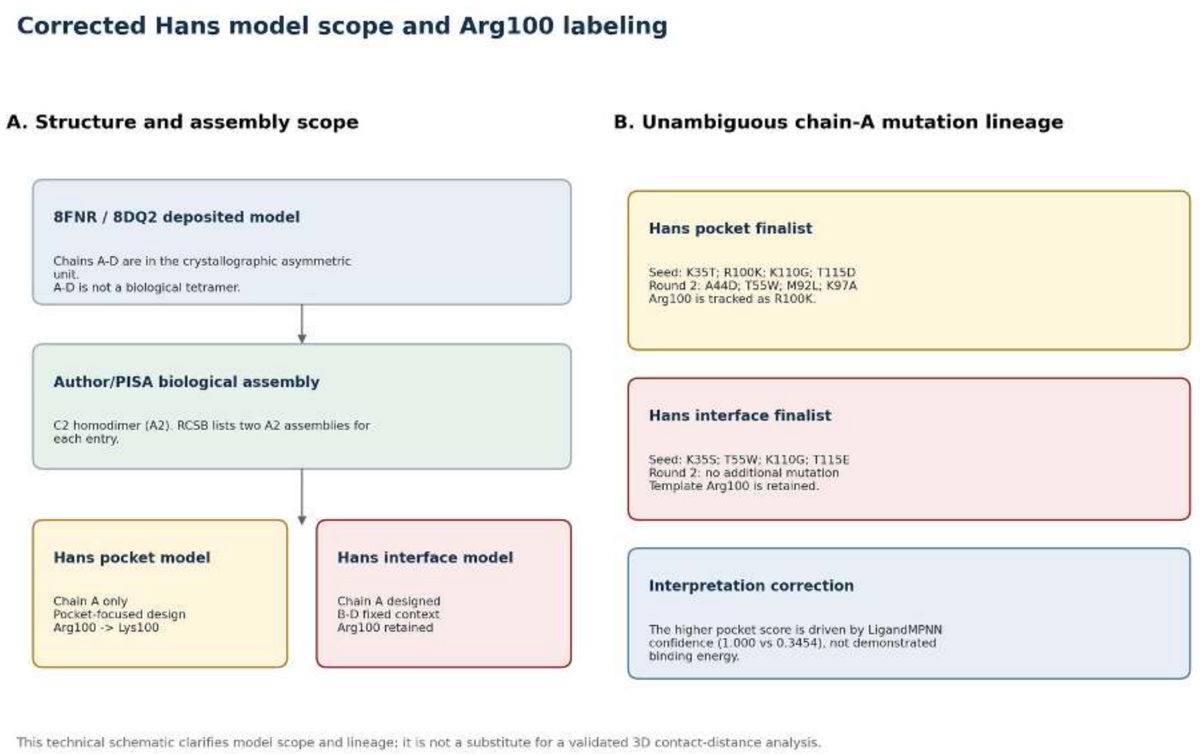

Figure TC3. Corrected Hans model scope and Arg100 labeling. The diagram distinguishes the crystallographic A-D context from A2 biological assemblies and explicitly tracks chain-A Arg100/Lys100.

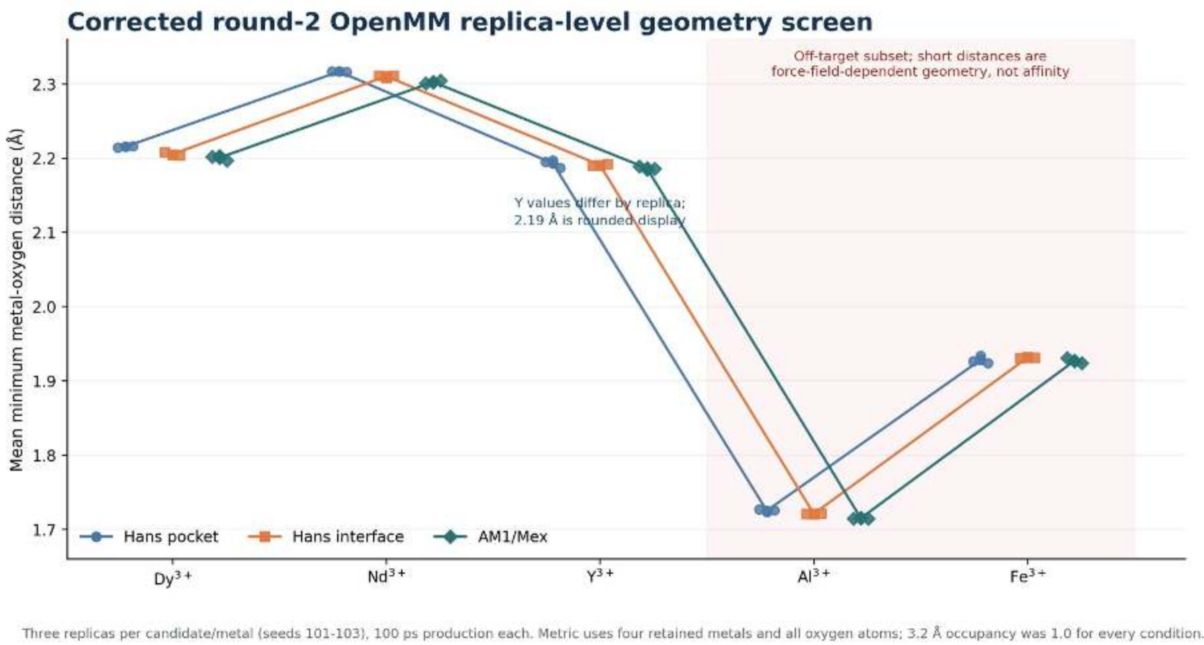

Figure TC4. Corrected round-2 OpenMM geometry screen. Every point is a replica mean. The Al/Fe values are force-field-dependent geometry results and are not interpreted as affinity or selectivity.

Figure TC5. Corrected model-quality prioritization. The Composite Quality Index is dimensionless and must not be labeled as Delta G, free energy, or thermodynamic affinity.

4. Corrected tables and exact technical settings

4.1 Design-mask reconciliation

Role	Memberships
First-shell	26
Second-sphere	16
Interface	10
Protected	18
Unique inventory	66
Membership sum	70
Overlap excess	4

Source: design_mask_candidates.csv; overlap detail is in mask_overlaps.csv.

4.2 MPNN and Rosetta specification

Component	Exact setting	Interpretation limit
ProteinMPNN round 1	20 sequences/target at temperatures 0.10 and 0.15; seed 37; batch size 1	Generative sequence sampling; not deterministic in the mathematical sense despite fixed seed.
ProteinMPNN round 2	20 sequences/target at temperatures 0.10 and 0.15; seed 37; batch size 1	Same fixed-seed sampling settings.
LigandMPNN rounds 1/2	Temperature 0.10; seed 37; 2 designs per candidate; one batch; side-chain and ligand context enabled	Model confidence is not a probability of thermodynamic binding.
Rosetta round 2	score_jd2; nstruct=1; seed 37; fa_elec weight=0; waters retained; auto metal setup; rank within topology	Score-only triage. No relaxation, no structural-realism acceptance cutoff, and no cross-topology score comparison.

Sources: recorded run_command.txt files; rosetta_round2_candidate_ranking.csv; repository scoring code.

4.3 OpenMM protocol and metric

Component	Exact setting	Interpretation limit
OpenMM round 2	298 K; 1 atm; 2 fs; friction 1 ps^-1; 100 ps production; 3 replicas, seeds 101-103; 500-step reporting	Short reduced rescreen; not a binding free-energy calculation.
OpenMM metric	Mean over frames of the minimum distance from each retained metal atom to any oxygen; 3.2 Å inner-sphere cutoff	Occupancy=1.0 at this cutoff does not establish selectivity or experimentally persistent binding.
Force field	amber19-all.xml + amber19/opc3.xml; generic published OPC3-compatible 12-6-4 metal family	LanM-chelator-tuned numeric parameters are absent; Al/Fe short distances require sensitivity analysis.

Sources: simulation_config.yaml, system_build_log.txt, and 45 screening_log.txt files under results/round2_openmm_screening/.

4.4 Generic OPC3-compatible 12-6-4 metal parameters

Metal	Charge	Rmin/2 (Å)	epsilon (kcal/mol)	C4 (kcal/mol Å^4)
Dy	3	1.632	0.0962022	183
Nd	3	1.712	0.1464093	184
Y	3	1.626	0.09289608	192
Al	3	1.361	0.01031847	363
Fe	3	1.455	0.02662782	429

Source: config/metal_parameter_values.yaml. The repository does not contain chelator-tuned numeric coefficients.

4.5 OpenMM candidate means

Candidate	Dy	Nd	Y	Al	Fe
Hans pocket	2.215 ± 0.0007	2.316 ± 0.0004	2.193 ± 0.0055	1.725 ± 0.0017	1.928 ± 0.0050
Hans interface	2.205 ± 0.0023	2.310 ± 0.0017	2.190 ± 0.0013	1.721 ± 0.0005	1.931 ± 0.0006
AM1/Mex	2.200 ± 0.0032	2.302 ± 0.0021	2.186 ± 0.0022	1.715 ± 0.0006	1.927 ± 0.0034

Values are mean ± sample SD of three replica-level means in Å. Full six-decimal data are in openmm_replicates.csv and openmm_summary.csv.

The near-identical Y values are expected from the common Y(III) parameter model and similar first-shell coordination. The underlying nine replica means range from 2.183940 to 2.196893 Å; the repeated '2.19 Å' statement was only rounded display.

4.6 Dimensionless model-quality index

Rank	Candidate	Ligand confidence	Overall confidence	CQI
1	Hans pocket	1.0000	0.3399	0.801970
2	Hans interface	0.3454	0.3303	0.340870
3	AM1/Mex	0.2213	0.3023	0.245600

CQI = 0.7 x ligand confidence + 0.3 x overall confidence. It is dimensionless and has no thermodynamic interpretation.

5. Structure and model-scope correction

PDB	Deposited model	Biological assembly	Technical use
8FNR	A,B,C,D; Dy: A=4, B=3, C=3, D=4 (14 total)	A2 homodimer (C2); two author/PISA assemblies are listed	Pocket model uses chain A; interface model keeps A-D crystallographic context
8DQ2	A,B,C,D; La: 3 per chain (12 total)	A2 homodimer (C2); author/PISA; gel-filtration evidence for assembly 1	Reference only; do not call A-D the biological tetramer
8FNS	A; Nd: 4	Monomeric reference in repository workflow	AM1/Mex chain-A model
6MI5	X; Y: 3	Monomeric reference in repository workflow	AM1/Mex reference

Primary structure sources: https://www.rcsb.org/structure/8FNR and https://www.rcsb.org/structure/8DQ2; repository inventory: results/tables/template_chain_summary.csv.

The deposited metal counts are crystallographic inventories, not numbers of equivalent functional sites. Any interface conclusion should identify the specific biological assembly or explicitly state that A-D was retained as crystallographic context.

6. Quantum evidence gap and submission gate

Submission gate after prototype execution

Do not present the atomic-Dy prototype figures, occupations, SOC values, or H2 CASPT2 reference as repository-reproduced LanM results. Manuscript Figures 6-7 remain gated until project-specific Dy and Nd cluster inputs and full outputs are added and independently regenerated.

Required artifact	Minimum content	Status	Affected output
Dy CASSCF input and full output	Coordinates, charge/multiplicity, Hamiltonian/ECP, basis sets, CAS(9,7), roots, convergence, occupations	LanM absent; atomic prototype added	Dy active-space claims; Figures 6-7
Nd CASSCF input and full output	Coordinates, charge/multiplicity, Hamiltonian/ECP, basis sets, CAS(3,7), roots, convergence, occupations	Absent	Nd active-space claims; Figures 6-7
Dy/Nd CASPT2 inputs and outputs	State count, imaginary and IPEA shifts, intruder-state diagnostics, energies and convergence	LanM absent; H2 reference added	CASPT2 comparison
SOC calculation files	SOC method, states mixed, spin-free roots, matrices/energies, convergence	LanM absent; atomic-Dy prototype added	Electronic-state ordering and anisotropy claims
Reduced-cluster coordinate and definition files	Atom/residue inventory, caps, constraints, charge/multiplicity, retained waters, cluster size	Absent	Environmental-scope claims

Required artifact	Minimum content	Status	Affected output
Natural occupation tables for every root	Seven active-orbital occupations per root with orbital identities	LanM absent; atomic-Dy prototype added	Active-space appropriateness
Orbital image source data	Molden/gbw or equivalent files plus orbital indices and plotting settings	LanM absent; atomic-Dy prototype added	Figure 7

Updated repository audit: genuine OpenMolcas prototype inputs, logs, tables, and selected orbital/anisotropy artifacts are now included. Project-specific Dy/Nd LanM production inputs and full outputs remain unavailable.

7. Repository correction set

Path	Purpose
README.md	Replaces the complete/reproducible and Delta-G language with a technical-correction notice and accurate limitations.
docs/technical_corrections/README.md	Human-readable scope, reviewer mapping, and regeneration instructions.
docs/technical_corrections/SOURCE_DATA_GAPS.md	Explicit quantum and manuscript-only actions that remain open.
docs/technical_corrections/LanM_Technical_Corrections_Report.{docx,pdf}	Shareable Word and PDF report.
results/technical_corrections/figures/*	Five corrected technical figures.
results/technical_corrections/tables/*	Machine-readable corrected data and provenance.
scripts/build_technical_corrections.py	Deterministic rebuild of tables, figures, report, and manifest.
requirements-technical-corrections.txt	Pinned reporting dependencies.
docs/technical_corrections/ LanM_Updated_Technical_Corrections_Report.{docx,pdf}	Updated shareable report including the prototype addendum and revised submission gate.
docs/technical_corrections/ OPENMOLCAS_PROTOTYPE_STATUS.md	Scope, completion evidence, and the project-specific evidence boundary.
results/openmolcas_prototype/*	Prototype figures, tables, successful logs, provenance, and selected source artifacts.
inputs/openmolcas_prototype/*; scripts/openmolcas_prototype/*	Exact prototype inputs and extraction/rerun scripts.

The manuscript file and its figures are intentionally not edited by this repository correction set.

8. Package manifest

The v2.0 bundle includes this updated Word report and PDF, seven distinct figures (five technical corrections and two prototype plots), the original twelve corrected/source-audit CSV tables, two updated status tables, eight prototype CSV tables plus one extraction JSON file, the complete validated prototype ZIP, repository-update files, manifests, and SHA-256 checksums.

Recommended share note

This v2.0 package resolves the technical, terminology, sampling, model-scope, and classical-data presentation corrections supported by the repository and adds authentic, independently repeated OpenMolcas prototype evidence. The prototype is method-validation evidence only; project-specific quantum claims and manuscript Figures 6-7 remain open.

9. Genuine OpenMolcas prototype execution update

Scope control

The following outputs are authentic OpenMolcas calculations and were independently checked. They validate an execution and extraction workflow, not the manuscript's Dy-LanM or Nd-LanM models. The two plots are prototype figures OM-P1 and OM-P2; they must not be renamed as manuscript Figures 6 or 7.

OpenMolcas v26.06 was built and run locally. A standard H2 reference exercised RASSCF and CASPT2, and an atomic Dy(III) workflow derived from official additional test 321 exercised CAS(9,7) RASSCF, RASSI spin-orbit coupling, and SINGLE_ANISO. The Dy workflow was repeated from a clean work directory and reproduced all extracted values at printed precision.

9.1 Software and execution record

Item	Recorded value
OpenMolcas	v26.06; source commit 8355057f32d65706a35996b5ab07cac2962bb728
Driver / compiler	pymolcas py2.32; GNU Fortran 13.3.0; Release build
Linear algebra / parallel mode	Internal BLAS/LAPACK; serial, one process and one thread
Dy integral treatment	Explicit RICD with CDTH 1.0D-6; DKH scalar relativity and AMFI integrals
Completion markers	_RC_ALL_IS_WELL_ for required stages and driver Happy landing

Exact build and input-change records are included under OpenMolcas_Prototype_Evidence and the repository prototype directories.

9.2 H2 RASSCF/CASPT2 reference validation

Method	Expected (Eh)	Observed (Eh)	Difference (Eh)	Within 1e-8
SCF	-1.122765460009	-1.122765460000	9.000e-12	TRUE
RASSCF	-1.141134852262	-1.141134852300	-3.800e-11	TRUE
CASPT2	-1.143847618496	-1.143847618500	-4.000e-12	TRUE

Source: unedited h2_console.log and h2_reference_validation.csv. All three energy differences are far below 1e-8 Eh.

9.3 Atomic Dy(III) CAS(9,7) workflow

Component	Prototype setting	Interpretation boundary
Model	Dy at origin with official test point-charge field	No protein, ligand shell, waters, MD frame, or reduced LanM cluster
Basis / relativity	ANO-RCC 7s6p4d2f1g; DKH; AMFI	Recorded only for this prototype
RASSCF	CAS(9,7), sextet, 11 state-averaged roots	Not a project-specific Dy-LanM active-space validation
SOC / anisotropy	RASSI mixing all 11 roots; 66 SOC states; SINGLE_ANISO	Not project-specific electronic-state ordering or anisotropy
Repeat	Two clean RICD runs; 89.29 s and 88.64 s	Numerical reproducibility of this test only

Source: dy3_ricd_core_console.log, dy3_ricd_repeat_console.log, and the supplied prototype input.

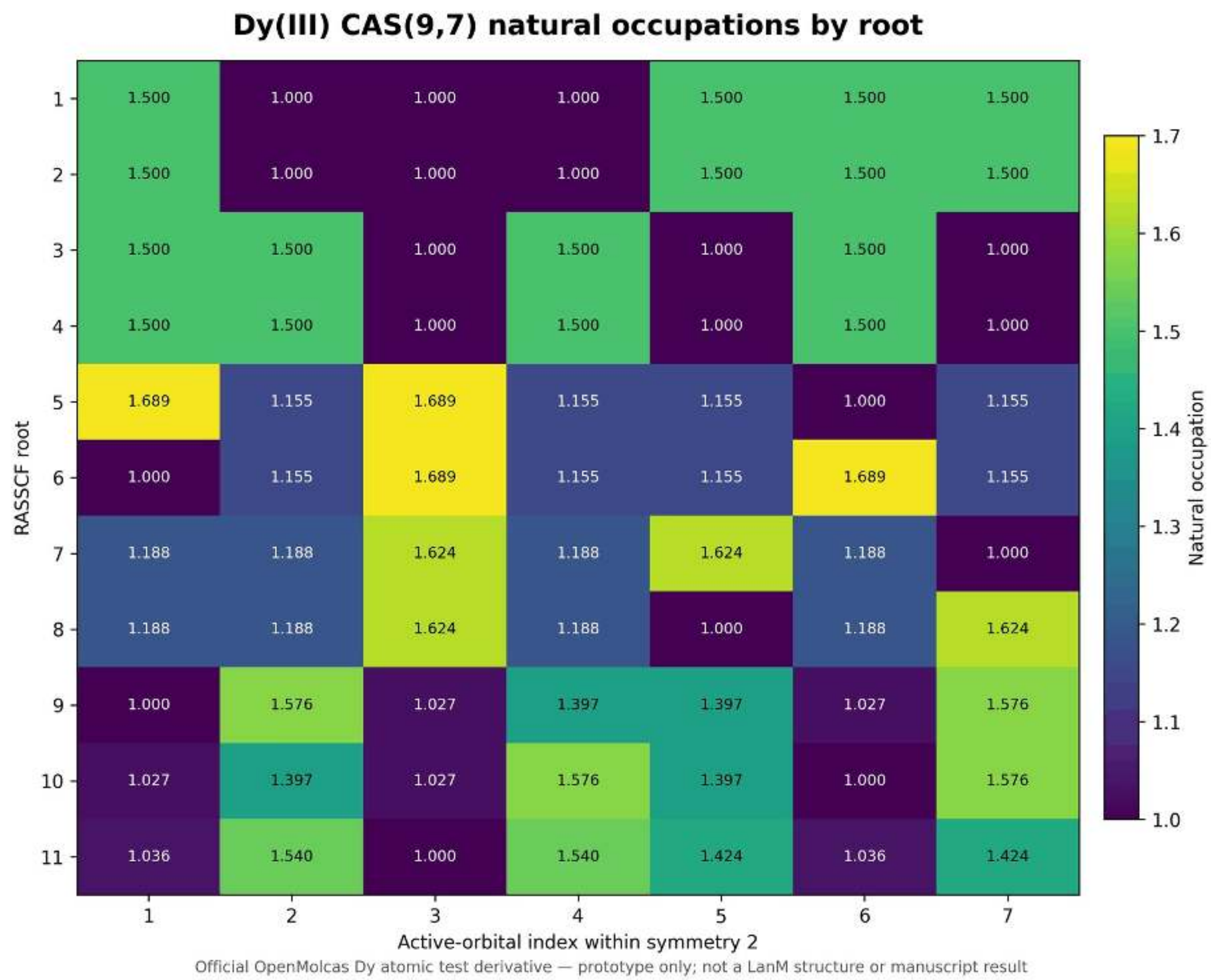

Figure OM-P1. Natural occupations for all 11 roots of the official atomic-Dy CAS(9,7) prototype. This is exact prototype output, not a Dy-LanM result and not manuscript Figure 7.

Each root has seven printed active-orbital occupations. Printed occupation sums are 9.000000 for roots 1-10 and 8.999998 for root 11 because of six-decimal rounding. The complete numeric table is supplied as dy3_natural_occupations_all_roots.csv.

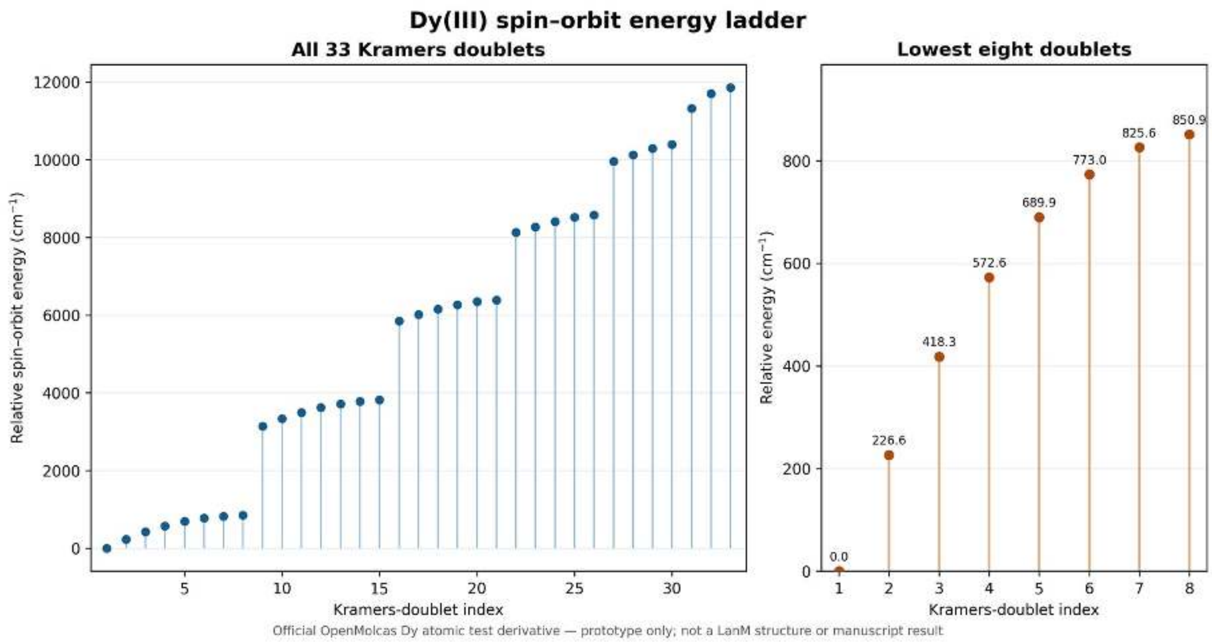

Figure OM-P2. Kramers-doublet energy ladder from the official atomic-Dy prototype RASSI/SOC calculation. This is exact prototype output, not a Dy-LanM state ordering and not manuscript Figure 6.

The 66 spin-orbit states form 33 Kramers pairs at printed precision. The lowest eight doublets occur at 0.0000, 226.5908, 418.3032, 572.5900, 689.9061, 773.0012, 825.5790, and 850.9258 cm-1. The idealized prototype ground doublet has principal values gX = 0.0000000, gY = 0.0000000, and gZ = 20.0115965.

9.4 Independent-repeat validation

Quantity	Maximum absolute difference	Identical at printed precision
RASSCF root energies (Eh)	0	TRUE
natural occupations	0	TRUE
spin-orbit relative energies (cm-1)	0	TRUE
ground-doublet g values	0	TRUE

Both complete successful Dy console logs are included; failed/partial conventional-integral attempts remain separated as diagnostics in the complete prototype ZIP.

9.5 What the prototype establishes

Question	Answer
Does an authentic OpenMolcas executable complete RASSCF/CASPT2 reference work?	Yes.
Does an atomic Dy CAS(9,7) -> SOC -> SINGLE_ANISO workflow complete and repeat?	Yes, twice with explicit RICD.
Are inputs, logs, tables, plots, and source artifacts supplied?	Yes, for the prototype.
Does this reproduce Dy-LanM or Nd-LanM?	No.

Question	Answer
Does this validate manuscript Figures 6-7 or project-specific CASPT2 ordering?	No.
Does this close reduced-cluster/environment sensitivity?	No.

10. Updated quantum evidence gate

Reviewer-submission gate

The technical correction package and the prototype run are complete and auditable. The primary manuscript's project-specific quantum goal is not achieved: Dy-LanM and Nd-LanM production calculations, CASPT2 comparison, SOC interpretation, reduced-cluster sensitivity, and authentic regeneration of Figures 6-7 still require project coordinates/definitions and full production outputs.

Evidence area	Added in v2.0	Project-specific status	Submission use
Dy CAS(9,7) occupations	Atomic-Dy prototype, all 11 roots	LanM absent	Method/execution evidence only
CASPT2	H2 reference validation	Dy/Nd LanM absent	Installation/reference evidence only
SOC / SINGLE_ANISO	Atomic-Dy prototype and repeat	LanM absent	Workflow evidence only
Orbital source data	Prototype RasOrb/Molden artifacts	LanM absent	Prototype traceability only
Reduced cluster	None	Absent	Cannot support environment claims
Figures 6-7	No project regeneration	Open	Do not claim corrected/reproduced

11. Repository and package changes

Change	Location / result
Updated technical report	Word and PDF under docs/technical_corrections/ and in the submission ZIP
Corrected technical figures	TC1-TC5 retained under results/technical_corrections/figures/
Updated correction tables	Revised reviewer and quantum-evidence status CSVs added without deleting the original audit tables
Prototype figures and tables	Added under results/openmolcas_prototype/ with explicit non-manuscript labels
Authentic source evidence	Inputs, successful logs, provenance, scripts, and selected orbital/anisotropy artifacts added
Integrity	SHA-256 manifests supplied for the package and repository subset

Technical conclusion

The targeted technical-correction result is achieved: the classical correction set is documented, figures and tables are packaged, the OpenMolcas execution path is authentically demonstrated, and repository artifacts are prepared. This is not equivalent to achieving the primary manuscript's project-specific quantum result.